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# MILITARY SPECIFICATON

## PLASTIC SHEET, ACRYLIC, MODIFIED, LAMINATED

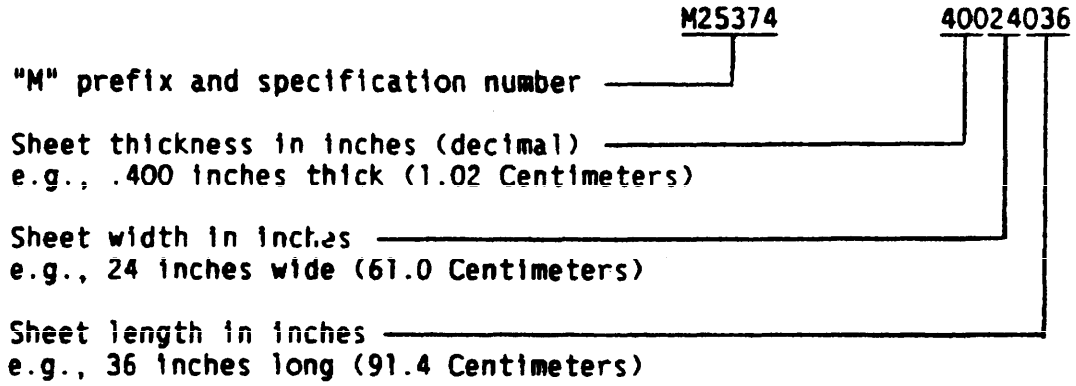
This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for laminated, modified acrylic sheet.

1.2 Classification. The laminated plastic sheet shall be of the dimensions (see 3.3) and following thicknesses, as specified (see 3.3.2 and 6.2.1).

1.3 Part identification. The sheets furnished under this specification shall be identified by part numbers consisting of the "M" prefix and specification number followed by an eight digit dash number consisting of sheet thickness in inches (three digits), sheet width in inches (two digits), and sheet length in inches (three digits) as shown in the following example:




Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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## 2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| NN-P-530 | - | Plywood, Flat Panel. |
| TT-N-95 | - | Naphtha, Aliphatic. |
| TT-W-572 | - | Wood Preservative, Water Repellent. |
| PPP-B-585 | - | Box, Wood, Wirebound. |
| PPP-B-591 | - | Boxes, Shipping, Fiberboard, Wood-Cleated. |
| PPP-B-601 | - | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - | Box, Wood, Nailed and Lock-Corner. |

MILITARY

| | | |
|---|---|---|
| MIL-P-8184 | - | Plastic Sheet, Acrylic, Modified. |
| MIL-L-10547 | - | Liner, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible. |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - | Marking for Shipment and Storage |

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ANSI/ASTM D 256 | - Impact Resistance of Plastics and Electrical Insulating Materials, Test Method of. |
| ANSI/ASTM D 637 | - Surface Irregularities of Flat Transparent Plastic Sheets, Test Method for. |
| ASTM D 648 | - Deflection Temperature of Plastics Under Flexural Load, Test Method for. |



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| | | |
|---|---|---|
| ASTM D 905 | - | Strength Properties of Adhesive Bonds in Shear by Compression Loading. |
| ANSI/ASTM D 1003 | - | Haze and Luminous Transmittance of Transparent Plastics, Test Method for. |
| ANSI/ASTM D 3029 | - | Impact Resistance of Rigid Plastic Sheeting on Parts by Means of a Tup (Falling Weight), Test Method for. |
| ASTM D 3951 | - | Packaging, Commercial |
| ASTM G 26 | - | Light-Exposure Apparatus (Xenon - Arc Type) With and Without Water for Exposure of Non-Metallic Materials, Recommended Practice for Operating. |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

## 3. REQUIREMENTS

3.1 Qualification. The plastic laminated sheets furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Laminate construction.

3.2.1 Face sheets. The face sheets shall be prepared from modified acrylic sheet material conforming to MIL-P-8184, Finish A. The material shall be a product listed on QPL-8184.

3.2.2 Interlayer composition. The interlayer material shall be polyvinyl butyral (PVB) plasticized with 37.5 $\pm$ 2.0 parts by weight of dibutyl sebacate (DBS) per 100 parts of polyvinyl butyral resin.

3.2.2.1 Purity of plasticizer. The dibutyl sebacate plasticizer used in the interlayer shall have a purity of at least 96 percent by weight and shall contain not more than 2.0 percent by weight of triethylene glycol di-2 ethyl butyrate (3GH).



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3.3 Dimensions and tolerances.

3.3.1 Surface dimensions. The dimensions of the sheets shall be as specified by the applicable drawings or specifications. Unless otherwise specified in the contract or order, tolerances of ± 0.125 inch (0.318 cm) will be allowed on the linear dimensions at 23.0 ± 1.1°C (73.5° ± 2°F) when the maximum dimension is less than 40 inches (1 m) and ± 0.375 inch (0.953 cm) when the maximum dimension is 40 inches (1 m) or more.

3.3.2 Thickness. The nominal thickness and configuration of the laminate sheet shall conform to Table I. The actual thickness of the sheet at any point, excluding a 2 inch (5 cm) margin around the edges, shall be within ± 10 percent of the nominal thickness for material having a maximum dimension of less than 40 inches (1 m) and shall be ± 15 percent for sheets having a maximum dimension of 40 inches (1 m) or more.

3.4 Optical uniformity.

3.4.1 Minor optical defects. The total number of minor optical defects (see 4.6.1.1) in the material shall not exceed a limit determined by dividing the area of the sheet, measured in square feet, by two. Minor defects include any embedded particles, blemishes, bubbles, or scratches which reduce visibility through the plastic. Defects within 2 inches (5 cm) of the edge of the sheet shall be disregarded, unless specified as an optically critical area. Lint, hairs and dust particles shall be considered blemishes if they individually affect an area greater than that of a 0.20 inch (0.5 cm) diameter circle. Blemishes which do not individually reduce visibility through the plastic shall be disregarded, unless they are grouped to form an objectionable pattern or appear in any critical vision area specified in the purchase contract.

3.4.2 Angular displacement. When tested as specified in 4.6.1.2, the angular displacement of the material shall be no greater than the values specified in Table I corresponding to the appropriate nominal thickness, excluding a 2 inch (5 cm) margin around the edges of the sheet, unless the acquisition activity specifies one or more edges to be optically critical.

3.4.3 Optical distortion. The optical distortion value of the material shall be not less than 12 inches (30.5 cm) excluding a 2 inch (5 cm) margin around the edges of the sheet, unless the acquisition activity specifies one or more edges to be optically critical.

3.5 Luminous transmittance. The luminous transmittance of the material, as received and after exposure, shall conform to Table I, when tested as specified in 4.6.2.

3.6 Haze. The haze in the material, when determined as specified in 4.6.2, shall be not greater than 4.0 percent, when measured on an area free of scratches in the as received, after accelerated weathering, and after Florida exposure.

3.7 Physical properties. The physical properties of the laminated plastic shall conform to Table II, when tested as specified in 4.6.3.3, 4.6.4.2, 4.6.5.1 and 4.6.5.2.



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3.8 Fracture resistance at low temperature. After testing as specified in 4.6.3.1, the material shall not fracture (break) into two or more pieces, shall not expose any bare interlayer and any delamination in any face material greater than 0.125 inch (0.31 cm) for qualification samples. There shall be no cracking of the laminate.

3.9 Falling ball impact. After impact by the falling ball as specified in 4.6.3.2, the material shall not break into two or more individual pieces. Small fragments (spalling) of the face material may separate from the specimen at the location diametrically opposite the point of impact but no portion of the interlayer material shall be exposed. The interlayer surface at point of impact shall be covered with particles of tightly adhered face material. There shall not be any sign of delamination beyond a one inch (2.5 cm) diameter circle directly opposite the point of impact, and no delamination greater than 0.125 inch (0.32 cm) inside the circle.

3.10 High temperature stability. After subjecting the material to high temperature as specified in 4.6.4.2, the specimen shall show no signs of bubbling, discoloration, or other indications of interlayer instability. In addition, the luminous transmittance shall be no lower than the as received material values specified in Table I, for the appropriate thickness. Haze shall not exceed 4.0 percent. Defects within 0.25 inch (.63 cm) of the edge of the specimen shall be disregarded.

3.11 Resistance to thermal shock. After exposure to thermal shock as specified in 4.6.4.3, the material shall show no evidence of delamination, cracking, crazing, or minor optical defects which were not present in the as received material, excluding a 0.25 inch (0.63 cm) margin around the edge of the sheet.

3.12 Formability. Formability shall be determined as specified in 4.6.4.4. The material shall be suitable for forming into hemispheric shapes with an outside diameter of 10 inches (25.4 cm) and a draw of at least 4.5 inches (11.4 cm).

3.13 Accelerated weathering. After exposure to accelerated weathering as specified in 4.6.6.1, the laminated plastic material shall show no crazing, cracking, surface instability, interlayer instability or other defects which significantly reduce visibility through the material which were not present in the as received material, excluding a 0.125 inch (0.32 cm) border around the edges. The values for luminous transmittance and haze shall be as prescribed in 3.5 and 3.6, respectively.

3.14 Resistance to Florida exposure. After exposure in southern Florida as specified in 4.6.6.2, the laminated plastic sheet shall show no crazing, cracking, delamination, surface instability, interlayer instability, or other defects which significantly reduce visibility through the material, excluding a 0.125 inch (0.32 cm) border around the edges. The laminate shall conform to the requirements specified in Table I, and to the requirements for haze and fracture resistance at low temperature specified in 3.6 and 3.8, respectively.



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3.15 Identification of product. Each laminated plastic sheet shall be indelibly marked on one corner by the laminator to designate the laminate correctly. This marking shall be on the interlayer material and be marked in such a way as to be legible after the lamination. The marking shall include the specification number, the day, month, and year of lamination, the manufacturer's name or trademark, type of acrylic face sheets, a code letter identifying the plasticizer used in the butyral interlayer, a code letter for interlayer supplier, and a code letter for the adhesive used in preparing the laminate. The identification marking shall consist of the following:

Part Identification Number
ABC-6-14-4-XY-12-Z2UE

Where:

ABC = Manufacturer's initials
6 = Month of lamination
14 = Day of lamination
4 = Last digit of calendar year of lamination
X = Symbol for interlayer manufacturer
Y = Symbol for interlayer plasticizer
12 = Piece number on the day produced
Z2 = Symbol for surface sheets
U = For UVA sheets
E = Symbol for lamination adhesive

3.15.1 Approval of marking code. Each manufacturer, when requesting initial product qualification approval (see 4.3.2), shall submit samples of the marking codes they intend to use. This procedure shall also be followed whenever a change in the materials is requested. Questions concerning marking codes should be referred to the qualifying activity (see 4.3.2).

3.16 Instruction sheet. An instruction sheet shall be furnished in each shipping container (see 5.2). Specific information or reference shall be made to the limitations of the material and the necessary precautions to be observed in handling, storing, cutting, drilling, machining, forming, bending, cementing, abrading, polishing and cleaning. The description of all compounds, materials and equipment mentioned therein shall be given in sufficient detail to permit nonproprietary acquisition, using Government specifications when available. Instruction sheets shall be approved by the acquisition activity (see 4.3.3).

3.17 Workmanship. Workmanship shall be in accordance with high-grade practice for this type of product. The finished sheet shall be free from bubbles, striae and other defects that would render the material unfit for the purpose of viewing objects through it. Since this specification does not apply to formed, molded or fabricated parts, workmanship beyond the production of polished plane sheet materials is not included.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except where otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 Qualification inspection. The qualification inspection shall consist of the tests specified in Table III.

4.3.1 Qualification test samples.

4.3.1.1 Sheet sample. The qualification sample shall consist of at least 20 square feet of laminated plastic sheet 0.500 inch (1.27 cm) thick. Individual sheets shall be a minimum size of 36 by 36 inches (0.91 by 0.91 meters). A portion of the sample shall be used by the manufacturer to prepare the specimens required in 4.3.1.2 with the remaining sheet material submitted to the qualifying laboratory.

4.3.1.2 Test specimens samples. A portion of the sheet samples shall be used by the manufacturer to prepare the test specimens specified in Figures 1 and 2. Preparation of these specimens may require the sample to be machined and polished as specified in the applicable test.

4.3.2 Forwarding of qualification samples. The qualification samples from 4.3.1.1 and 4.3.1.2 shall be forwarded to the Commander, Naval Air Development Center, Warminster, PA 18974, Attention: Aircraft and Crew Systems Technical Directorate (Code 6063). The samples shall be plainly and durably marked with the following information:

Sample for qualification tests
PLASTIC SHEET, ACRYLIC, MODIFIED, LAMINATED
Name of manufacturer
Plant address (where manufactured)
Submitted by (name) (date) for qualification testing in accordance with the requirements of MIL-P-25374B under (reference authorizing letter) (see 6.3).



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4.3.3 Manufacturer's data. Two copies of the manufacturer's test report shall be submitted with the samples of 4.3.1. The report shall contain numerical test data, where applicable, showing that the material submitted for the qualification inspection conforms to the requirements of this specification. In addition, the manufacturer shall certify to the requirements of 3.2 through 3.2.2.1. Two copies of the manufacturer's instruction sheet (see 3.16) shall be furnished.

4.3.4 Retention of qualification. In order to retain qualification of a product approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that the manufacturer's product complies with the requirements of this specification. The time of periodic verification by certification shall be in two-year intervals from the date or original qualification. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specifications requirements.

4.4 Quality conformance inspection.

4.4.1 Lot formation. Unless otherwise specified, a lot shall consist of the total number of sheets forming part of one contract or order, submitted for inspection at one time.

4.4.2 Sampling.

4.4.2.1 Laminated plastic sheets. All the laminated plastic sheets in the lot shall be visually examined and dimensionally checked. Reject all defective units.

4.4.2.2 Packaging inspection. A quantity of shipping containers fully prepared for delivery, just prior to closure, shall be selected at random in accordance with MIL-STD-105, Inspection Level S-2 and an Acceptable Quality Level (AQL) of 4.0 percent defective. The lot size for purposes of this inspection shall be the total number of shipping containers. The sample unit shall be one shipping container.

4.4.2.3 Physical and mechanical properties. A random sample of laminated plastic sheets consisting of 10 percent of the lot, shall be selected for determining the physical and mechanical properties specified in 4.4.3.2. Reject the lot if any laminated plastic sheet or specimen fails to comply with the requirements of this specification.

4.4.3 Inspection methods.

4.4.3.1 Visual examination.

4.4.3.1.1 Laminated plastic sheets. Each laminated plastic sheet shall be visually examined for conformance to this specification for dimension and tolerances (3.3), minor optical defects (3.4.1), identification of product (3.15) and workmanship (3.17).



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4.4.3.1.2 Packaging. The samples selected shall be examined to the requirements of Table IV and section 5 of this specification. In addition, shipping containers fully prepared for delivery shall be examined for closure defects.

4.4.3.2 Physical and mechanical properties. Each of the laminated plastic sheets, shall be tested for minor optical defects (3.4.1), angular displacement (3.4.2) and optical distortion (3.4.3). After determining the aforementioned properties, half of the selected sample sheets from the lot shall be tested for luminous transmittance (3.5) and haze (3.6) followed by high temperature stability (3.10). Specimens for the heat stability test shall be a 4 inch (10cm) right triangle cut from one corner of the sheet.

4.5 Inspection conditions.

4.5.1 Standard conditions. Standard conditions shall be 25° ± 1°C (77° ± 2°F) and a relative humidity of 50 ± 5 percent. Unless otherwise specified, all tests and examinations shall be conducted at standard conditions.

4.5.2 Reporting of test results. Unless otherwise specified in the applicable test method, all results shall be reported as the average of the number of specimens being tested. Each individual value shall also be reported.

4.6 Test methods.

4.6.1 Optical uniformity.

4.6.1.1 Minor optical defects. The as received laminated sheets shall be examined visually for defects which are allowed only to the extent specified in 3.4.1.

4.6.1.2 Angular displacement and optical distortion. The angular displacement of each laminated sheet shall be determined in accordance with ASTM D 637. The obtained results shall conform to 3.4.2 and 3.4.3.

4.6.2 Luminous transmittance and haze. Luminous transmittance and haze shall be determined in accordance with ASTM D 1003, Method A. Material subjected to accelerated weathering or Florida exposure shall be cleaned prior to testing by immersing momentarily in distilled water, then removing the surface water.

4.6.3 Impact tests.

4.6.3.1 Fracture resistance at low temperature. Two 8 by 8 inch (20 by 20 cm) specimens shall be cooled in air to -18° ± 2°C (0° ± 4°F) for a time sufficient to insure a uniform temperature throughout the specimen. While at the same temperature, the laminate shall be placed on a rigid flat surface which has a one inch (2.54 cm) diameter cut-out in its center. The specimen shall be positioned so the geometric center is above the cut out of the surface. A 2 pound steel ball shall be dropped from a height of 10 feet directly onto the center point of the specimen. Two specimens shall be impacted for qualification, one for quality conformance (see 4.4.2.3) to determine conformance to 3.8.



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4.6.3.2 Falling ball impact. The falling ball impact test shall be conducted in accordance with ASTM D 3029, Method F, Tup A, total weight 2 pounds (0.91 kg), with one drop from a height of 20 feet (6.1 m). Conformance to 3.9 shall be noted.

4.6.3.3 Izod impact strength. Izod impact strength of 5 specimens shall be determined in accordance with Method A of ASTM D 256. Dimensions shall be as specified in the method except the specimen shall not be notched and the thickness of the laminate shall be considered the specimen thickness. Impact strength, ft lbs/inch (J/M), shall be determined by dividing energy expended in individual test by actual thickness of the specimen to determine conformance to Table II.

4.6.4 Thermal testing.

4.6.4.1 High temperature stability. Two specimens, 4 by 6 inches (100 by 150 mm), shall be suspended in a circulating air oven at 135° ± 2°C (275° ± 4°F) for a period of time in minutes equal to the thickness of one surface sheet (Table I) times 200. Exposure times will vary from 25 through 75 minutes. Upon completion of the exposure, the specimen shall be gradually cooled to standard conditions, then visually examined and tested for luminous transmittance and haze. The specimens shall conform to 3.10. One specimen shall be tested for quality conformance testing (see 4.4.2.3).

4.6.4.2 Heat distortion temperature. At least two conditioned specimens shall be tested as specified in ASTM D 648, Procedure 6 (a) to determine conformance to Table I.

4.6.4.3 Thermal shock. Two specimens, 4 by 6 inches (100 by 150 mm) shall be suspended in naphtha conforming to TT-N-95 at a temperature of -40° ± 5°C (-40° ± 10F) for a period of time, in minutes, equal to the thickness of one surface sheet times 400. Exposure times will vary from 50 through 150 minutes. When completed, the specimens shall be removed from the liquid, wiped clean with a soft cloth and immediately suspended in a liquid medium at 100° ± 2°C (212° ± 4°F) for the same time as the low temperature exposure. Upon completion, the specimens shall be removed from the liquid, wiped clean as before, and examined for conformance to 3.11.

4.6.4.4 Formability. Two specimens shall be formed to determine conformance to 3.12. Forming conditions shall be in accordance with the manufacturer's instructions. Material with a thickness up to 0.500 inch shall be tested in the as-cast form. Material of greater thickness shall be machined to 0.500 inch prior to testing.

4.6.5 Interlayer tests. (See Table II).

4.6.5.1 Shear strength of lamination. Five specimens shall be prepared as shown in Figure 3. The shear strength of the laminated plastic shall be determined in accordance with ASTM D 905. The specimen shall be mounted in the shearing apparatus in such a manner that the dynamic loading arm applies pressure to the face of the specimen which is extended beyond the interlayer while the interlayer and other face of the laminated specimen rest on the static plate of the shearing apparatus. The load shall be applied at such a rate that the cross-head closure rate is 0.20 ± 0.02 inch (0.51 ± 0.05 cm) per



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minute. The shear strength of the laminated material shall be reported as the average of the five specimens, calculated in psi by dividing the maximum load by the area of the interlayer which is common to both face sheets in the specimen.

4.6.5.1.1 After Florida exposure. The shear strength of material after subjection to Florida exposure shall be determined in accordance with the method described in 4.6.5.1 with the following exceptions:

a. Three specimens shall be sheared on the face sheet which was nearer to the sun during exposure.

b. Three specimens shall be sheared on the face sheet which was farther from the sun during exposure.

4.6.5.2 Interlayer peeling rate. Five specimens shall be prepared as shown in Figure 4. One face sheet shall be removed from each specimen in such a manner that the bond between the interlayer and the other face sheet is not disturbed. The interlayer shall be peeled down approximately 1 inch (2.5 cm) from one end of the remaining face sheet. This specimen shall be mounted in a jig as shown in Figure 5. A total weight of 10 pounds (4.5 kg) shall be attached to the free end of the interlayer and the time shall be noted and the line shall be made to determine the peeling rate of the interlayer from the face sheet. The peeling rate, over at least 3 inches (7.6 cm) of the specimen, shall be averaged and the average reported in minutes per inch (min/cm).

4.6.5.2.1 After Florida exposure. The interlayer peeling rate of material after subjection to Florida exposure shall be determined in accordance with the method described in 4.6.5.2 with the following exceptions:

a. Three specimens shall incorporate the face sheet that was farther from the sun during exposure.

b. Three specimens shall incorporate the face sheet that was nearer the sun during exposure.

4.6.6 Environmental exposure.

4.6.6.1 Accelerated weathering. Accelerated weathering shall be conducted in accordance with Method A of ASTM G 26 for 240 hours. Each specimen shall be visually examined to the requirements of 3.13 and then subjected to the luminous transmittance and haze tests, 4.6.2.

4.6.6.2 Florida exposure. Unprotected laminated panels (for qualification only) shall be exposed outdoors in southern Florida for a period of 6 months. The specimens shall be mounted as to be unrestrained during exposure. The specimens shall be supported at the edges and shall not be backed. They shall be mounted at a 45 degree angle with the horizontal facing south and both surfaces of the specimen shall be open to air circulation. At least one panel, 18 by 22 inches (46 by 56 cm), shall be exposed. After exposure, test specimens shall be cut as specified in Figure 2 and then subjected to Table II tests, haze (3.6) and fracture resistance at low temperature (3.8) to determine conformance to 3.14.



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4.7 Packaging inspection. One sample pack shall be selected at random from each lot and be inspected to determine conformance with Section 5.

5. PACKAGING

5.1 Preservation - packaging. Preservation - packaging shall consist of a protective covering on both sides of the plastic sheet. The protective covering shall be a suitable adhered paper or film that can be readily removed without injury or damage to the plastic surface. The covering shall adequately protect the surfaces from scratches or damage during shipment.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2.1). Containers shall be suitable for maintaining the plastic sheet in its original condition. All internal loads shall be supported to avoid damage to surfaces by cleats, etc. An instruction sheet shall be included in each individual exterior container (see 3.16). All plastic sheets shall be packed as far as practicable by size and thickness.

5.2.1 Level A. Plastic sheets, packaged as specified in 5.1, shall be packed for shipment in overseas type containers conforming to PPP-B-585, Class 3, PPP-B-591, PPP-B-601 or PPP-B-621, Class 2, Style 2. Plywood shall conform to NN-P-530, Type I or II, Grade 4 of Group A. Plywood shall be treated in accordance with TT-W-572. Containers conforming to PPP-B-591 or PPP-B-601 shall be modified to the extent that solid wood ends and sides, in lieu of the cleated type, shall be used. Wirebound boxes conforming to PPP-B-585 shall be provided with fiberboard liners having a minimum Mullen test of not less than 275 pounds. All cleats shall be flush and containers shall be designed in a manner which will insure even weight distribution over the entire bearing surface when the sheets are stored. Containers shall be closed and strapped in accordance with the applicable container specification and appendix thereto. Containers conforming to PPP-B-585, PPP-B-591 or PPP-B-621 shall be provided with a case liner fabricated and sealed in accordance with MIL-L-10547. The gross weight of the shipping container, when packed for shipment, shall not exceed approximately 200 pounds, except when the weight of a single packed sheet exceeds this limitation.

5.2.2 Level B. Plastic sheets, packaged as specified in 5.1, shall be packed for shipment in domestic type containers conforming to PPP-B-585, PPP-B-591, PPP-B-601 or PPP-B-621, Class 1, Style 2. Containers conforming to PPP-B-591 or PPP-B-601 shall be modified to the extent that solid wood ends and sides in lieu of cleated type shall be used. Wirebound boxes conforming to PPP-B-585 shall be provided with fiberboard liners having a minimum Mullen test of not less than 275 pounds. All cleats shall be flush and containers shall be designed in a manner which will insure even weight distribution over the entire bearing surface when the sheets are stored. Containers shall be closed and strapped in accordance with the applicable container specification and appendix thereto. The gross weight of the shipping container, when packed for shipment, shall not exceed 500 pounds, except when the weight of a single packed sheet exceeds this limitation.

5.2.3 Commercial. Plastic sheets, packaged as specified in 5.1, shall be packed for shipment in exterior type shipping containers conforming to the requirements of ASTM D 3951. Containers shall meet Uniform Classification Rules of other common carriers as applicable to the mode of transportation.



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5.3 Marking of shipment.

5.3.1 Individual sheets. The protective covering (see 5.1) of each plastic sheet shall be distinctly marked, at intervals of 1 foot, with the same markings specified in 3.15. In addition, each protective covering shall be marked with unit markings in accordance with MIL-STD-129.

5.3.2 Shipping containers. Each shipping container shall be marked in accordance with MIL-STD-129, and with any special markings below or in the contractor order.

Stock number (NSN) or other identification number as specified in the purchase order or contract.
PLASTIC SHEET, ACRYLIC, MODIFIED, LAMINATED
Part Identification Number.
Quantity.
Size and thickness.
Color of plastic sheet (when required).
Date of manufacture (month and year).
Stored items shall be issued by their date of manufacture (i.e., older items shall be issued first).
Contract or order number.
Name of manufacturer.
Name of contractor (if different from manufacturer).

NOTE: The contractor shall enter the NSN specified in the purchase document or as furnished by the acquiring activity. When the NSN is not provided or available from the acquiring activity, leave space therefore and enter the number or other identification as provided by the acquiring activity.

6. NOTES

6.1 Intended use. This material is primarily intended for transparent areas on aircraft where a material with optical, formability, outdoor weathering, shatter resistance and heat resistance properties are required.

6.1.1 Optical purposes. In case material conforming to this specification is used for specialized optical purposes, more rigid optical requirements may be specified in the detail product specification. In this event, the requirements of the detail specification prevail.

6.1.2 Drawing references. Reference of this specification on drawings or in product specifications may be made in order to specify the flat material to be used in the fabrication of a part. It must be clearly understood that the requirements herein apply to the flat material prior to forming or fabrication and not necessarily to the material in its final form.

6.2 Ordering data.



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6.2.1 Acquisition documents. Acquisition document should specify the following:

a. Quantity

b. Size of sheets

c. Thickness of sheets

d. MIL-P-25374B

e. Level of packing

f. Surface sheet thickness

g. Which corners shall be used in inspection tests

h. Location of interlayer butt joints upon request of laminator

i. Optically critical areas

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL 25374) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Washington, DC 20361; however, information pertaining to qualification of products and letter of authorization for submittal of sample may be obtained from the Commander, Naval Air Development Center, Warminster, PA 18974, Attn: Aircraft and Crew Systems Technical Directorate (Code 6063).

6.4 Changes from previous issue. Asterisks are not used in this specification to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MR
Navy - AS
Air Force - 20

Review activities:
DLA - GS

Preparing activity:
Navy - AS

(Project 9330-B058)



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Table I. Nominal thickness and configuration.

| Laminate nominal thickness (cm) | Individual surface sheet nominal thickness, inch (cm) | Interlayer nominal thickness, inch (cm) | (Minimum luminous transmittance) | | | Angular displacement (minutes) |
|---|---|---|---|---|---|---|
| | | | Original (%) | After accelerated weathering (%) | After Florida exposure (%) | |
| 0.400 (1.016) | 0.125 (.381) | 0.150 (.381) | 90 | 89 | 88 | 10 |
| .450 (1.143) | .150 (.381) | .150 (.381) | 89 | 88 | 87 | 11 |
| .500 (1.270) | .150 (.381) | .200 (.508) | 89 | 88 | 87 | 12 |
| .600 (1.524) | .200 (.508) | .200 (.508) | 88 | 87 | 86 | 14 |
| .950 (2.413) | .375 (.953) | .200 (.508) | 87 | 86 | 85 | 20 |





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Table II. Physical properties. 1/

| Characteristics | As received | | After Florida exposure | | Test paragraph |
|---|---|---|---|---|---|
| | Minimum individual value | Average | Minimum individual value | Average | |
| Izod impact strength, ft lbs/in (J/M) | 7(140) | - | 7(140) | - | 4.6.3.3 |
| Heat distortion Temperature | 85°C (185°F) | - | - | - | 4.6.4.2 |
| Shear strength, psi (KPa) | 450(3100) | 500(3500) | 450(3100) | 500(3500) | 4.6.5.1 |
| Interlayer peeling rate, inch/minute (cm/min) | 2(5) | - | 1(2.5) | - | 4.6.5.2 |

1/ A dash "-" in the value column indicates a property not determined.



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Table III. Qualification inspection.

| Characteristics | Paragraph | |
|---|---|---|
| | Requirement | Test |
| Face sheets | 3.2.1 | 4.4.3 |
| Interlayer material | 3.2.2 | 4.4.3 |
| Plasticizer | 3.2.2.1 | 4.4.3 |
| Dimensions | 3.3 through 3.3.2 | 4.4.3.1.1 |
| Optical uniformity: | 3.4 | - |
| Minor optical defects | 3.4.1 | 4.4.3.1.1 |
| Angular displacement | 3.4.2 | 4.6.1.2 |
| Optical distortion | 3.4.3 | 4.6.1.2 |
| Luminous transmittance | 3.5 | 4.6.2 |
| Resistance to thermal shock | 3.11 | 4.6.4.3 |
| Formability | 3.12 | 4.6.4.4 |
| Interlayer tests: | - | 4.6.5 |
| Shear strength of lamination | Table II | 4.6.5.1 |
| Interlayer peeling rate | Table II | 4.6.5.2 |
| Environmental exposure: | - | 4.6.6 |
| Accelerated weathering | 3.13 | 4.6.6.1 |
| Florida exposure | 3.14 | 4.6.6.2 |
| Identification of product | 3.15 | 4.4.3.1.1 |
| Instruction sheet | 3.16 | 4.4.3.1.1 |
| Workmanship | 3.17 | 4.4.3.1.1 |



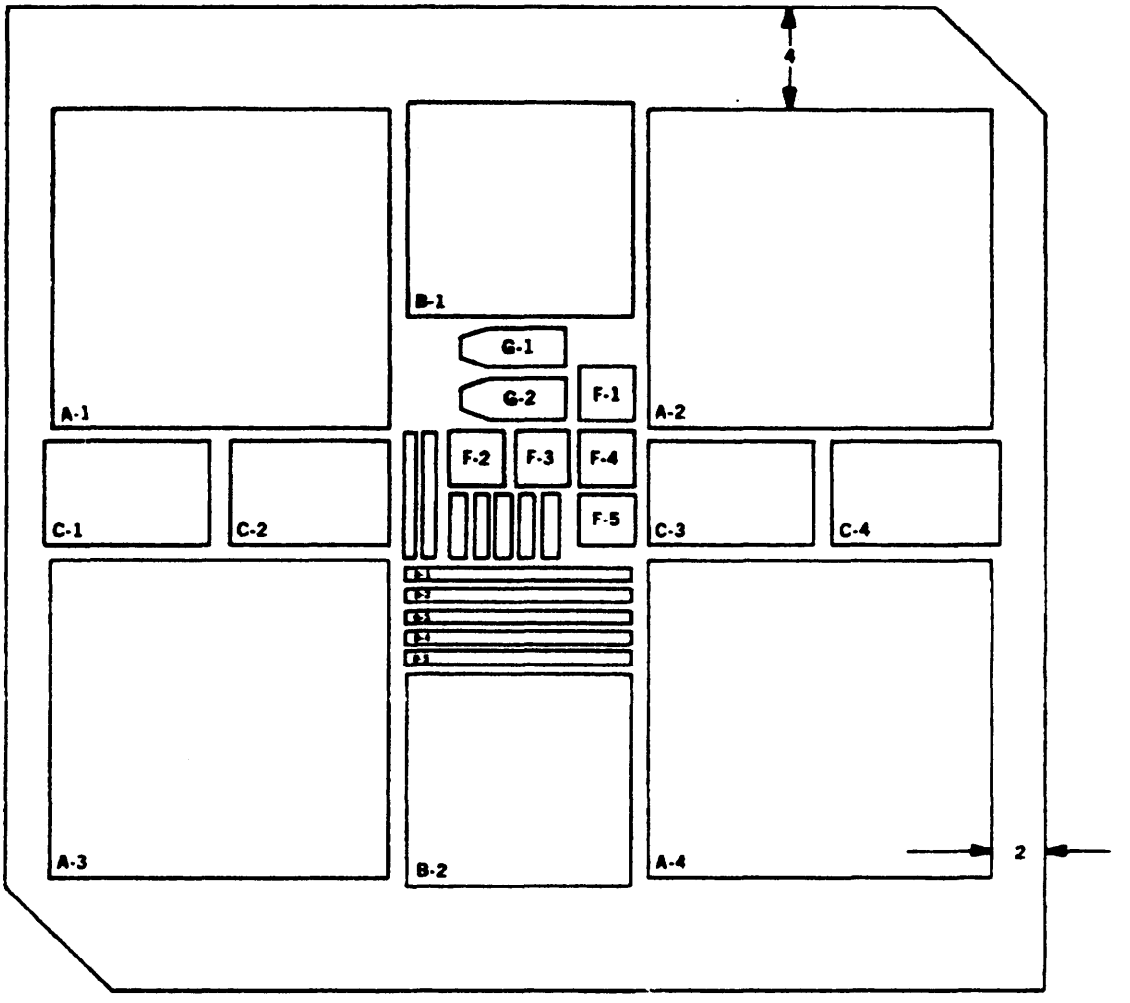
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Table IV. List of defects for packaging.

| Examine | Defect |
|---|---|
| ging | Individual sheets not packaged as specified. |
| ng | Packing material not as specified.<br>Not in accordance with contract requirements. |
| | Container not as specified, closure not accomplished by specified or required methods or materials. |
| | Inadequate application of components, such as incomplete closures of case liner, container flaps, loose or inadequate strappings, bulged or distorted containers. |
| uction sheet | Missing or not as specified (see 3.16). |
| | Less than specified or indicated quantity. |
| t | Gross or net weight exceeds specified requirements. |
| rgs | Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements (see 5.3). |



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DIMENSIONS IN INCHES (CM) TOLERANCES
± 0.1 INCH ( 25 CM), UNLESS OTHERWISE SPECIFIED

**SPECIMEN LAYOUT**

A FALLING BALL IMPACT 12 BY 12 (30 5 BY 30.5)

B LOW TEMPERATURE FRACTURE 8 BY 8 (20 BY 20)

C 1 & 3 THERMAL SHOCK 4 BY 6 (10 BY 15)
2 & 4 HIGH TEMP STABILITY 4 BY 6 (10 BY 15)

D PEELING RATE 8 BY 0.500 ± 0.005 (20 BY 1.25 ± 01)

E IZOD IMPACT 2 5 BY 0.500 ± 0 005 (6 35 BY 1 25 ± 01)

F SHEAR STRENGTH. SEE FIGURE 3

G ACCELERATED WEATHERING

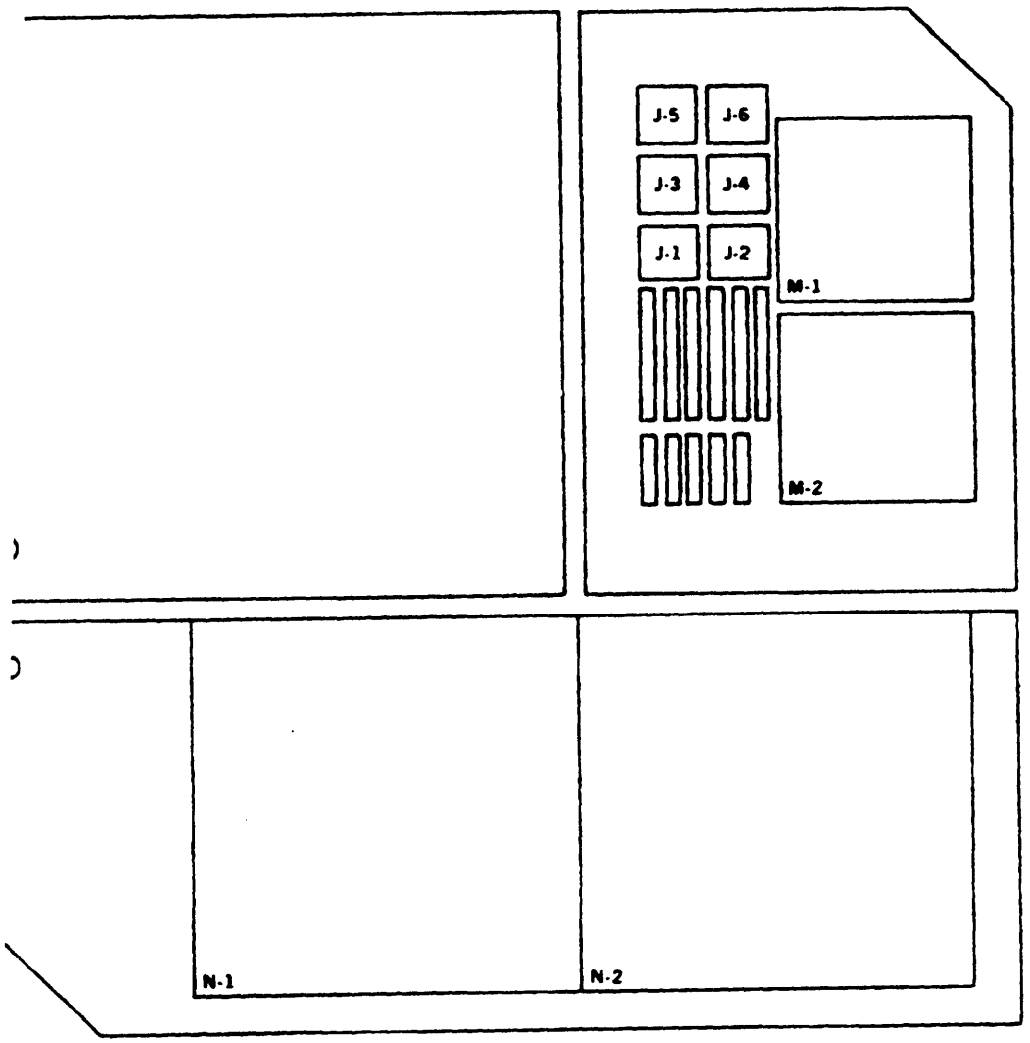
H FLEXURAL DEFORMATION TEMP
5 BY 0.500 ± 0.005 (12 7 BY 1.25 ± 01)

EDGES OF ALL SPECIMENS EXCEPT
A, B, & C ARE TO BE MACHINED

FIGURE 1. Specimen layout.



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DIMENSIONS IN INCHES (CM) TOLERANCES
± 0.1 INCH (.25 CM) UNLESS OTHERWISE SPECIFIED

**SPECIMEN LAYOUT**

IGHT SECTION 18 BY 22 (45.7 BY 55.9) TO BE
) AS A WHOLE FOR 6 MONTHS
H FLORIDA

.R STRENGTH SEE FIGURE 3

ING RATE 8 BY 0.500 ± 0.005 (20 BY 1.25 ± .01)

IMPACT 2.5 BY 0.005 ± 0.005 (6.35 BY 1.25 ± .01)

/ TEMPERATURE FRACTURE 7 BY 7 (18 BY 18)

LOWER SECTION 37 BY 15 (94 BY 38), NOT
WEATHERED

N FORMABILITY 14 BY 14 (36 BY 36)

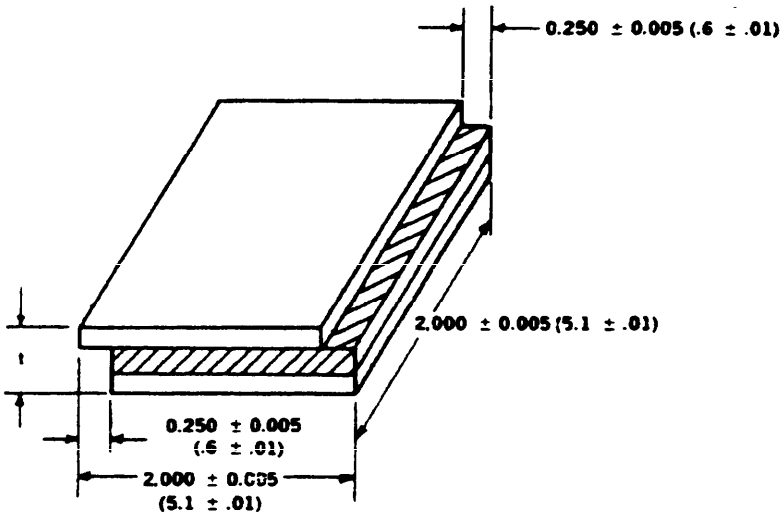
C EXTRA UNWEATHERED MATERIAL

MACHINE EDGES OF SPECIMENS J, K & L

FIGURE 2. Specimen layout.

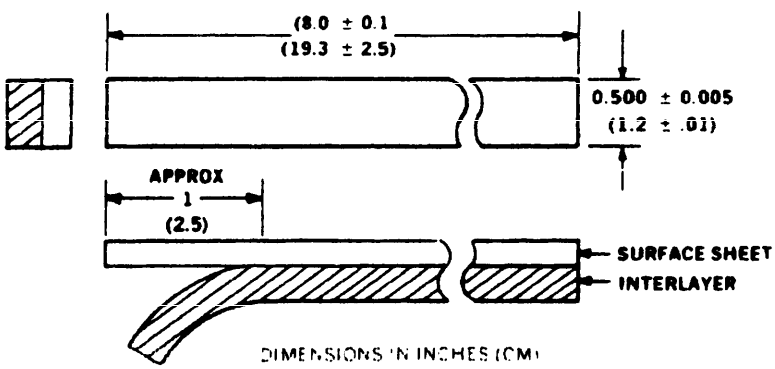


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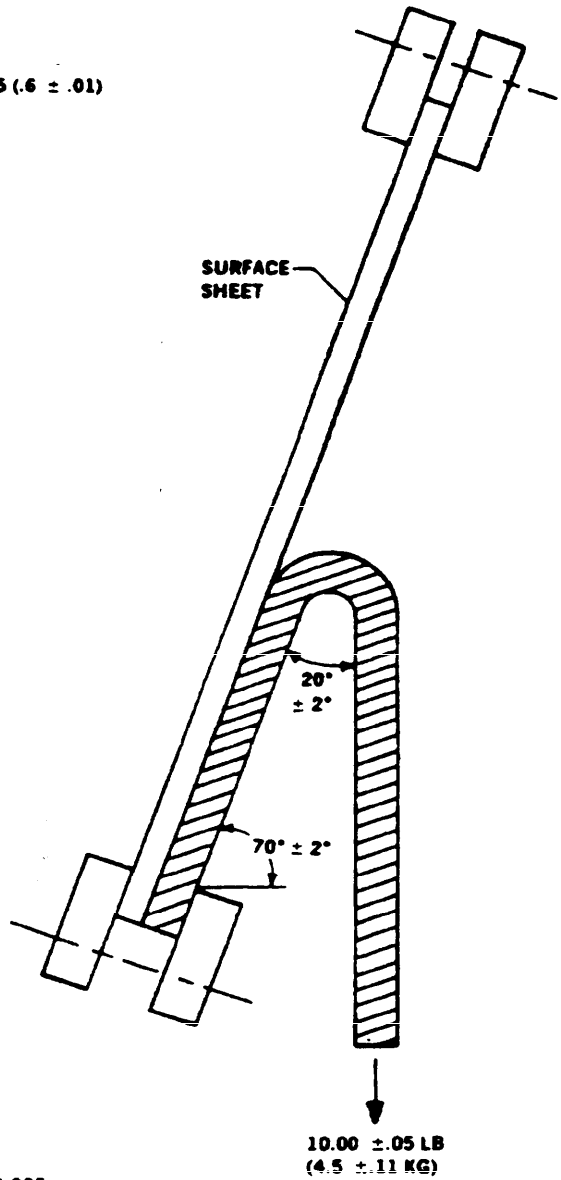

DIMENSIONS IN INCHES (CM)

FIGURE 3. Shear strength specimen.




DIMENSIONS IN INCHES (CM)

FIGURE 4. Peeling rate specimen.




DIMENSIONS IN INCHES (CM)

FIGURE 5. Test position of peeling rate specimen.


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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| MENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| -25374B | Plastic Sheet, Acrylic, Modified, Laminated |

| E OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| IESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

ILEM AREAS

agraph Number and Wording:

ecommended Wording:

ason/Rationale for Recommendation:

IARKS

| ME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| LING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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